S.E.IT
Scheme

Program Structure for Second Year of Information Technology

UNIVERSITY OF MUMBAI (With Effect from 2025-2026)

SEMESTER III

Course Code	Course Description	Teaching Scheme (Contact Hours)			Credit Assigned			
		Theory	Practical	Tutorial	Theory	Tutorial	Practical	Total Credits
2343111	Applied Mathematics Thinking-I	2	--	1-	2	1	--	3
2343112	Advance Data Structure & Analysis	3	–	--	3	--	--	3
2343113	Database Management System & Application	3	--	--	3	--	--	3
2343114	Automata Theory	3	--	--	3	--	--	3
OEC301	Open Elective	2#	--	--	2	--	--	2
2343115	ADSA Lab	--	2	--	--	--	1	1
2343116	SQL Lab	--	2	--	--	--	1	1
2343611	Mini-Project - Full Stack Java Programming	–	2*+2	–	–	–	2	2
2993511	Entrepreneurship Development	--	2*+2	---	--	--	2	2
2993512	Environmental Science	--	2*+2	--	--	--	2	2
Total		**13**	**16**	**01**	**13**	**01**	**08**	**22**

* Two hours of practical class to be conducted for full class as demo/discussion.

Theory / Tutorial 1 credit for 1 hour and Practical 1 credit for 2 hours

Institute shall offer a course for Open Elective from Science/Commerce/Management stream bucket provided by the University of Mumbai.

#Institute shall offer a course for MDM from other Engineering Boards.

Course Code	Course Description	Examination scheme							
		Internal Assessment Test (IAT)			End Sem. Exam Marks	End Sem. Exam Duration (Hrs)	Term Work (Tw)	Oral & Pract.	Total
		IAT-I	IAT-II	Total (IAT-I) + IAT-II)					
2343111	Applied Mathematics Thinking-I	20	20	40	60	2	25	--	125
2343112	Advance Data Structure & Analysis	20	20	40	60	2	--	--	100
2343113	Database Management System & Application	20	20	40	60	2	--	--	100
2343114	Automata Theory	20	20	40	60	2	--	--	100
OEC301	Open Elective	20	20	40	60	2	--	--	100
2343115	ADS Lab	--	--	--	--	--	25	25	50
2343116	SQL Lab	--	--	--	--	--	25	25	50
2343611	Mini-Project - Full Stack Java Programming	--	--	--	--	--	50	25	75
2993511	Entrepreneurship Development	--	--	--	--	--	50	--	50
2993512	Environmental Science for Engineers	--	--	--	--	--	50	--	50
Total		**100**	**100**	**200**	**300**	**10**	**225**	**75**	**800**

Program Structure for Second Year of Information Technology

UNIVERSITY OF MUMBAI (With Effect from 2025-2026)

SEMESTER IV

Course Code	Course Description	Teaching Scheme (Contact Hours)			Credit Assigned			
		Theory	Practical	Tutorial	Theory	Tutorial	Practical	Total Credits
2344111	Applied Mathematics Thinking-II	2	--	1	2	1	–	3
2344112	Operating System	3	–	--	3	–	–	3
2344113	Computer Network & Network Design	3	--	--	3	–	–	3
MDC401	Multidisciplinary minor	3	–	--	3	–	–	3
OEC401	Open Elective	2#	–	--	2	–	–	2
2344114	Unix Lab	–	2	–	–	–	1	1
2344115	Network Design Lab	–	2	–	–	–	1	1
MDL401	Multidisciplinary minor	–	2	–	–	–	1	1
2344411	Mini-Project -Programming Paradigm	--	2*+2	--	--	--	2	2
2994511	Business Model Development	–	2*+2	–	–	–	2	2
2994512	Design Thinking	–	2*+2	–	–	–	2	2
Total		**13**	**18**	**01**	**13**	**01**	**09**	**23**

* Two hours of practical class to be conducted for full class as demo/discussion.

Theory / Tutorial 1 credit for 1 hour and Practical 1 credit for 2 hours

Students must select course for Open Elective from Science/Commerce/Management stream bucket provided by the University of Mumbai.

#Institute shall offer a course for MDM from other Engineering Boards.

Course Code	Course Description	Examination scheme							
		Internal Assessment Test (IAT)			End Sem. Exam Marks	End Sem. Exam Duration (Hrs)	Term Work (Tw)	Oral & Pract.	Total
		IAT-I	IAT-II	Total (IAT-I) + IAT-II)					
2344111	Applied Mathematics Thinking-II	20	20	40	60	2	25	--	125
2344112	Operating System	20	20	40	60	2	--	--	100
2344113	Computer Network & Network Design	20	20	40	60	2	--	--	100
MDC401	Multidisciplinary minor	20	20	40	60	2	--	--	100
OEC401	Open Elective	20	20	40	60	2	--	--	100
2344114	Unix Lab	--	--	--	--	--	25	25	50
2344115	Network Design Lab	--	--	--	--	--	25	25	50
MDL401	Multidisciplinary minor	--	--	--	--	--	25	--	25
2344411	Mini-Project - Programming Paradigm	--	--	--	--	--	50	25	75
2994511	Business Model Development	--	--	--	--	--	50	--	50
2994512	Design Thinking	--	--	--	--	--	50	--	50
Total		**100**	**100**	**200**	**300**	**10**	**250**	**75**	**825**

Vertical – 1
Major

Sem. – III

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2343111	Applied Mathematics Thinking-I	02	-	01	02	-	01	03

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	IAT-I + IAT-II					
2343111	Applied Mathematics Thinking-I	20	20	40	60	2	25	--	125

Note: * One hour of tutorial class to be conducted for full class as practice/problem solving/discussion/theory.

Course Objectives: Students will be able to learn:

1. To familiarize with the Laplace Transform, Inverse Laplace Transform of various functions, and its applications.
2. To acquaint oneself with the concept of Fourier series, its complex form, and enhance problem-solving skills.
3. To familiarize the concept of complex variables, C-R equations with applications.
4. The fundamental knowledge of Trees, Graphs, etc.
5. To study the basic techniques of statistics, including correlation, regression, and curve fitting, for data analysis, Machine learning, and AI.
6. To understand some advanced topics of probability, random variables with their Distributions and expectations.

Course Outcomes:

Sr. No.	Course Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy
On successful completion, of course, learner/student will be able to:		
1	Apply the concept of Laplace transform to solve the real integrals in engineering problems.	L1, L2
2	Apply the concept of inverse Laplace transform of various functions in engineering problems.	L1, L2
3	Expand the periodic function by using Fourier series for real life problems and complex engineering problems.	L1, L2, L3
4	Find orthogonal trajectories and analytic function by using basic concepts of complex variable theory.	L1, L2, L3
5	Apply the concept of Correlation and Regression to the engineering problems in data science, machine learning and AI.	L2, L3
6	Illustrate understanding of the concepts of probability and expectation for getting the spread of the data and distribution of probabilities.	L1, L2

DETAILED SYLLABUS:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Applied Mathematics I, Applied Mathematics-II	01	--
I	**Laplace Transform**	Definition of Laplace transform, Condition of Existence of Laplace transform,	**06**	**CO1**

		Laplace Transform of Standard Functions likee^{at}, $sin(at)$, $cos(at)$, $sinh(at)$, $cosh\,(at)$and t^n, $n\geq 0$. Properties of Laplace Transform: Linearity, First Shifting Theorem, Change of scale Property, multiplication by *t*, Division by *t (division by t^2 not included),* Laplace Transform of derivatives (up to second derivative) and integrals (Properties without proof) Evaluation of real integrals by using Laplace Transformation. **Self-learning Topics:** Heaviside's Unit Step function, Second shifting theorem, Laplace Transform. Of Periodic functions, Dirac Delta Function.		
II	**Inverse Laplace Transform**	Inverse Laplace Transform, Linearity property, use of standard formulae to find inverse Laplace Transform, finding Inverse Laplace transform using derivatives, Partial fractions method to find inverse Laplace transform. (only up to s^3 in denominator) Inverse Laplace transform using Convolution theorem (without proof) Applications to solve initial and boundary value problems involving ordinary differential equations (up to 2nd order differential equation) **Self-learning Topics:** Partial fractions method to find inverse Laplace transform. (with s^4 in denominator), Inverse Laplace transform using derivatives and its properties.	**06**	**CO1, CO2**
III	**Fourier Series**	Dirichlet's conditions, Definition of Fourier series and Parseval's Identity (without proof) Fourier series of periodic function with period2 ⌜ and2*l*, Fourier series of even and odd functions (simple functions only, piecewise continuous function not to be included) Hal Frange Sine and Cosine Series. **Self-learning Topics:** Complex form of Fourier Series, orthogonal and orthonormal set of functions, Fourier Transform.	**05**	**CO3**
IV	**Complex Variables**	Analytic function, necessary and sufficient conditions for *f*(z) to be analytic (without proof), Cauchy-Riemann equations in cartesian coordinates (without proof, Polar form not included) Milne-Thomson method to determine analytic function *f*(z) when real (u)or imaginary part (v) is given Harmonic function, Harmonic conjugate, and orthogonal trajectories **Self-learning Topics:** Cauchy-Riemann equations in polar coordinates, conformal mapping, linear, bilinear mapping, cross ratio, fixed points, and standard transformations.	**05**	**CO4**
V	**Statistical Techniques**	Kar lPearson's Coefficient of correlation(r) Spearman's Rank correlation coefficient (R) (with repeated and non-repeated ranks) Lines of regression Fitting of first- and second-degree curves. **Self-learning Topics:** Covariance, fitting of exponential curve.	**04**	**CO5**
VI	**Probability**	Discrete and continuous random variable with probability distribution and probability density function. Expectation of random variable with mean, variance and standard deviation, moment generating function up to two moments.	**04**	**CO6**

		Self-learning topics: Total probability theorem, Bayes' theorem, Skewness and Kurtosis of distribution (data).		
Note: No questions will be asked in the end-semester exam from self-study topics. However, students are encouraged to explore these topics for a better understanding of the subject.				

Text Books:

1. Higher Engineering Mathematics, Dr. B.S. Grewal, Khanna Publication
2. Advanced Engineering Mathematics, Erwin Kreyszig, Wiley Eastern Limited.
3. Advanced Engineering Mathematics, R.K. Jainand S.R.K. Iyengar, Narosapublication,

References:

1. Complex Variables and Applications, Brown and Churchill, McGraw-Hill Education.
2. Probability, Statistics and Random Processes, T. Veerarajan, McGraw-Hill education.
3. Theory and Problems of Fourier Analysis, Murray Spiegel, Schaum'sOutlineSeries.
4. Higher Engineering Mathematics, H. K. Dass
5. Text book of Engineering Mathematics, N. P. Bali and Dr. Manish Goyal

Online References:

Sr. No.	Website Name
1.	https://www.nptel.ac.in
2	Higher Engineering Mathematics by H.K.Dass
3	Higher Engineering Mathematics by B.V. Raamana

List Of Tutorials :

Tutorial No	Tutorial Topic	Hours
1	Laplace Transform	1
2	Inverse Laplace Transform	1
3	Application Of Laplace Transform	1
4	Fourier Series (Full range)	1
5	Half Range Fourier Series	1
6	Complex Variables	1
7	Statistical Techniques	1
8	Probability	1

Term Work:

General Instructions:

1. Students must be encouraged to write at least 6 class tutorials on the entire syllabus.
2. A group of 4-6 students should be assigned a self-learning topic. Students should prepare a presentation/problem solving of 10-15 15minutes. This should be considered as a mini project in

Applied Mathematics. This project should be graded for 10 marks depending on the performance of the students.

The distribution of Term Work marks will be as follows–

1	Attendance (Theory and Tutorial)	05marks
2	Class Tutorials on entire syllabus	10marks
3	Mini project	10marks

Assessment:

Internal Assessment Test (IAT) for 20 marks each:

- IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

End Semester Theory Examination:

- **Question paper format**

- Question Paper will comprise a total of **six questions each carrying 15 marks Q.1** will be **compulsory** and should **cover the maximum contents of the syllabus**

- **Remaining questions** will be **mixed in nature** (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)

- A total of **four questions** need to be answered.